DIMUTHU WIJESIRIWARDENE

ROBOTICS & AUTONOMOUS SYSTEMS RESEARCHER

MOBILE ROBOTICS • PERCEPTION & CONTROL

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Location : Colombo, Sri Lanka

EDUCATION

- BEng (Hons) Electrical and Electronic Engineering — Curtin University, Perth, Australia — Jan 2016 - Jan 2020
  - Upper Second Class Honours
- GCE Advanced Level — Dharmapala Vidyalaya Pannipitiya, Colombo, Sri Lanka — Jan 2011 - Sep 2014
  - Physical Science Stream
    - Math, Physics & Chemistry

RESEARCH EXPERIENCE

- Graduate Research Assistant — Faculty of Engineering, University of Sri Jayewardenepura, Colombo, Sri Lanka — Oct 2021 - Jun 2022
- Undergraduate research assistant (Internship) — Sustainable Computing Research Lab (SCoRe Lab), University of Colombo School of Computing, Colombo, Sri Lanka — Nov 2017 - Jan 2018

PROFESSIONAL EXPERIENCE

- Application Engineer — Synopsys Inc. Colombo, Sri Lanka — Jun 2022 - Present
- Associate Software Engineer — Insharp Technologies Pvt. Ltd. Colombo, Sri Lanka — Aug 2020 - Oct 2021

ROBOTICS AND RESEARCH PROJECTS

- Fire Rescue Assisting Robot (Project Post)
  - Designed and developed an individual honors thesis investigating thermal perception for fire rescue robots
  - Developed a human detection system using a thermal sensor and computer vision techniques for smoke-filled environments for deployment on UGV/UAV robots
  - Final year individual research project at Curtin University (2019)
- NovasPlay - Omnidirectional Soccer robot (Demo Video)
  - Designed and developed an omnidirectional soccer robot with a custom dribbling mechanism
  - Modeled and fabricated the robot using Autodesk Fusion, laser-cut sheet metal and 3D-printed components
  - Achieved a 5-1 victory in the initial round of the Synopsys robotics competition (2024)
- Reverse Vending Machine (Demo Video)
  - Led the university side development of an industry collaborative automated Reverse Vending Machine project integrating a vision based identification system, embedded processing and automated sorting
  - Designed and simulated a 3D model of a reverse vending machine in Blender to aid prototype development
  - A research project at the University of Sri Jayewardenepura in collaboration with OREL Corporation (2022)
- Low-cost Time Domain Reflectometer (Project Post)
  - Designed and built a low-cost time-domain reflectometer for electric fence fault detection as part of the elephant-human conflict mitigation research project
  - Developed and experimentally validated the prototype through circuit simulation, pulse generation, oscilloscope based measurements of signal reflections - Research project at SCoRe Lab (2017)
- Low-cost Fence Energizer (Project Post)
  - Reverse engineered a GALLAGHER B180 Multi Power Fence Energizer through PCB trace analysis, continuity measurements and circuit reconstruction
  - Reconstructed the system in Multisim and supported the development of a lower-cost alternative as part of the elephant-human conflict mitigation research project at SCoRe Lab (2017)

- 3D Printed Quadruped Robot (Demo Video)
  - Developed and fabricated a 3D-printed quadruped robot integrating servo actuators and embedded control hardware for research into mobility over challenging terrain
  - Assembled and validated the platform through actuator testing and locomotion experiments for a planned beach waste collection research project at SCoRe Lab (2018)

- The Wind Turbine (Demo Video)
  - Designed, fabricated, assembled and evaluated a small-scale wind turbine prototype demonstrating mechanical to electrical energy conversion
  - Team member and presenter, Principles and Communication module project at the university (2016)

- The Automated Bedsheet (Demo Video)
  - Led the development from initial concept through mechanical design, fabrication and functional prototyping of a system for automated bedsheet deployment and wrinkle reduction
  - Designed and fabricated a custom stepper motor driver reducing prototype hardware cost by approximately 85%
  - Presented the project at the university engineering exhibition ("Young Engineering Expo", 2016)

ENGINEERING AND SOFTWARE PROJECTS

- Verdi - Debug and Verification Management Platform
  - Performed incoming bug analysis, root cause analysis and validation of new features and enhancements for customers such as Apple, Intel, NVIDIA, AMD
  - Developed test automation solutions using scripting and AI-Driven testing methodologies; contributed new regression tests to the regression suite, improving verification coverage and quality assurance processes
  - Served as the product validation owner of Verdi Transaction Debug component with quality assurance & signoff responsibilities
  - Worked with Synopsys VIP designs & UVM, performing tool specific testing - Project at Synopsys

- VCS - Verilog Compiled Simulator
  - Validated and debugged VCS Intelligent Coverage Optimization (ICO), an AI/ML driven technology for accelerating functional coverage closure in semiconductor verification
  - Validated integration of ICO with Verdi RDA to support identification of faulty code & traces - Project at Synopsys

- Euclide - Real-Time Hardware Development and Verification IDE
  - Validated new features & enhancements & performed RCA of customer reported issues - Project at Synopsys

- jTerm - Efficient Jira creation tool
  - Developed a tool for streamlined Jira creation with built-in support for screenshots and annotations via the in-house *Snipper* tool
  - Adopted by Synopsys engineering teams globally with 500+ Jiras created to date
  - Presented at the internal poster presentation event ("Purple Poster Event", 2025) at Synopsys

- jCrash - Streamlined crash reporting tool
  - Developed a command line tool to report application crashes instantly by executing a single command
  - Received the Synopsys Innovations Award & Presented at the event ("Purple Poster Event", 2024) at Synopsys

- gCron - Crobjob manager
  - Developed an intuitive interface for cron job management with natural-language scheduling, real-time previews and backup support; adopted by teams globally
  - Presented at the internal poster presentation event ("Purple Poster Event", 2026) at Synopsys

- Additional Engineering and Software Projects
  - weGen - Weekly Report Generator
    - Unified project status & weekly reporting into a single interface
  - PerfTest - Automated Performance Testing
    - Automated Performance testing across multiple customer designs
  - Cdiff - Customized Diff Tool
    - Developed a customized diff utility for efficient engineering workflows
  - TuTi - Virtual Classroom
    - Developed a full stack virtual classroom platform at Inshar Technologies
  - Retailz - Data Analytics Platform
    - Developed a full stack data analytics platform for supermarkets and suppliers

PUBLICATIONS

- Wijesiriwardene, D. and Thilakumara, R. "Human Detection for Fire Rescue Robots Using a Thermal Sensor and Hu Moment Invariants", Accepted for presentation at the *IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEEET), 2026.*

AWARDS, PRESENTATIONS AND RECOGNITIONS

- 2024 - Award Recipient, Synopsys Innovations Award (*jCrash*)
- 2026 - Presenter, Purple Poster Event (*gCron*)
- 2025 - Presenter, Purple Poster Event (*jTerm*)
- 2024 - Presenter, Purple Poster Event (*jCrash*)
- 2024 - Competitor, Synopsys RoboCup Competition (*NovasPlay*) - Demo Video
- 2016 - Presenter, Young Engineering Expo (*Automated Bedsheet*) - Demo Video

TECHNICAL SKILLS

- Robotics & Embedded Systems
  - Raspberry Pi, Arduino, PIC, MSP430, PLC
  - Sensors, Actuators
  - Robot Operating System (ROS 2)
  - Hardware Interfacing
  - I2C, SPI, UART, PWM
- Computer Vision, Simulation & Numerical Tools
  - OpenCV, Pandas
  - NumPy, SciPy
  - MATLAB
  - Godot Engine
- Electronics & PCB Design
  - Circuit Simulation: NI Multisim, Proteus
  - PCB Design: Eagle PCB
  - PCB Manufacturing: Toner-transfer Method
  - PCB Reverse Engineering
  - Oscilloscopes, Multimeters, Soldering
- Web & Software Development
  - Django, Laravel
  - HTML, CSS
  - JavaScript, Vue.js
  - Blockchain (Smart contracts, Solidity)
- Programming & Software
  - Python, C, Shell scripting (Csh, Bash), Tcl
  - Version Control (Git)
  - Linux (Ubuntu, Raspbian)
  - AI-assisted Development
  - Prompt Engineering
- CAD, Design & Fabrication
  - Autodesk Fusion, Blender, AutoCAD, SolidWorks
  - 3D-Printing (PLA, ABS, TPU, TPE), Sheet Metal Fabrication (laser cutting)
  - Mechanical Assembly; Power Tools
- Validation, Verification & Quality Assurance
  - Synopsys Verdi (Debugging)
  - Synopsys VCS (Simulation)
  - Root Cause Analysis, Jira
  - Selenium, Test Planning
  - AI-based Test Case Creation Agents
- Technical Communication & Media
  - Poster Presentations
  - Technical Demonstrations
  - Video Production
  - Video Editing (Adobe Premiere Pro)